Additional file 2. Early renewal of accountable care organization agreements to reset benchmarks in the Medicare Shared Savings Program: a preregistered retrospective cohort study

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All tables are generated from the committed results files by `scripts/make_tables.py` in the manuscript repository. Item and section numbers refer to the registered analysis plan (`PREREGISTRATION.md`, version 0.6).[1]

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eTable 1. Validation of the rebuilt benchmark against benchmarks CMS published for regular renewers

Check	ACOs, No.	Spearman ρ	Median absolute error, %	Slope	Intercept, $
Rebuilt 2022 benchmark, all	101	0.94	1.6	0.944	664
ACOs at the first phase-in step	62	0.93	2.0	0.917	886
ACOs at the second phase-in step	39	0.96	0.9	1.017	−47
Own-spending part (step 1) vs CMS benchmark-year spending	101	0.95	1.4	0.940	676

Validation sample: regular renewers whose 2022 benchmarks CMS publishes (101 of 111 eligible; the rest lack either the rebuilt or the published benchmark). Early renewers were excluded from the sample, and the 2022 results file was filtered as it was loaded. Registered pass rule: Spearman correlation at least 0.80 and median absolute percentage error at most 5%; calibration only if the slope of actual on rebuilt fell outside 0.9 to 1.1. Result: passed, no calibration. The error SD of the step-1 check sets the measurement-error SD of the own-spending component, $\sigma_v = 4.38$ pp.

eTable 2. Power analysis

Test	Spread used	Minimum detectable effect	Note
H1, savings rate	SD 4.9 pp (PY2016-PY2018)	2.7 pp	simulated power 80.4%
H2b, regional pull	SD 4.6 pp (stayers)	2.5 pp	0.55 SD
H2a, own-spending component	SD 3.3 pp (stayers)	1.8 pp	0.55 SD
H4, spending change	SD 6.9 log points (pre-period change)	3.8 log points	+3.9% or −3.7%
H3, regional pull	true log-odds difference 0.5 per 2 SD	detected in 17%	of 500 simulated datasets
H3, own-spending component	true log-odds difference 0.5 per 2 SD	detected in 0%	of 500 simulated datasets
H3, higher BASIC level	true log-odds difference 0.5 per 2 SD	detected in 14%	of 500 simulated datasets

Minimum detectable effects at 80% power, two-sided $\alpha = .05$, for 29 early renewers vs 277 stayers, computed after registration and committed before any confirmatory estimate; 1 log point = 0.01 in log spending. For H3, power was simulated only at a log-odds difference of 0.5; a predictor counts as detected when its 95% credible interval excludes zero.

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eTable 3. Robustness of the group comparisons

Analysis	Difference	95% interval	*P*
H2b regional pull			
Confirmatory	+3.4	1.7 to 5.4	<.001
Regression, cohort only (5.3)	+2.1	0.5 to 3.8	
Regression + track, level and size (item 1)	+2.2	0.7 to 3.8	
Difference in medians (item 2)	+2.8	0.1 to 5.2	<.001
2020-01-01 cohort only (item 3)	+3.7	1.8 to 5.5	.001
Unadjusted county spending (item 9)	+6.9	2.4 to 12.4	.002
H2a own-spending component			
Confirmatory	−2.1	−3.8 to −0.5	.003
Regression, cohort only (5.3)	−1.5	−2.8 to −0.3	
Regression + track, level and size (item 1)	−1.3	−2.6 to 0.0	
Difference in medians (item 2)[a]	−1.4	−3.2 to 0.0	.09
2020-01-01 cohort only (item 3)[a]	−1.8	−4.2 to 0.5	.04
BY2-BY3 only in step 1 (item 6)	−1.9	−3.7 to −0.2	.01
Complete cases (item 8)	−2.1	−3.9 to −0.5	.005
H1 savings rate, PY2020			
Confirmatory	+1.9	0.7 to 3.3	.01
Regression, cohort only (5.3)	+1.5	0.2 to 2.9	
Regression + track, level and size (item 1)	+1.2	−0.1 to 2.6	
Difference in medians (item 2)	+1.4	−0.1 to 3.3	.06
2020-01-01 cohort only (item 3)	+1.8	0.5 to 3.3	.03
Total expected gain			
Confirmatory	−0.5	−2.0 to 0.8	.33
Complete cases (item 8)	−0.5	−2.0 to 0.8	.33
Regional component			
Confirmatory	+1.6	0.9 to 2.2	<.001
Complete cases (item 8)	+1.6	0.9 to 2.2	<.001

Early renewers minus stayers, in the units of Table 2. Item numbers refer to the registered plan's robustness list (section 5.5); '5.3' is the registered covariate-adjusted estimate with cohort only. Regression rows are robust Bayesian linear regressions (Student-t errors on the standardized variable, Normal(0, 1) priors) of the variable on early renewal and cohort (5.3) or cohort, track, level and size (item 1), with 95% credible intervals and no P value; the robust likelihood downweights a few large values. Median rows give the bootstrap posterior mean of the difference in medians; their P

value tests the observed difference. [a]The P value and the 95% interval disagree about zero; the plan requires reporting both and noting the disagreement.

eTable 4. Robustness of the decision model (H3)

Analysis	Coefficient	95% interval	*P*
Regional pull			
Confirmatory	1.34	0.44 to 2.30	
Firth logistic regression (item 5)	1.33	0.25 to 2.54	.004
2020-01-01 cohort only (item 3)	1.15	−0.02 to 2.39	
BY2-BY3 only in step 1 (item 6)	1.33	0.42 to 2.26	
No measurement-error model (item 7)	1.22	0.34 to 2.20	
Complete cases (item 8)	1.34	0.38 to 2.32	
Unadjusted county spending (item 9)	0.75	−0.02 to 1.56	
Own-spending component			
Confirmatory	−0.05	−2.01 to 1.92	
Firth logistic regression (item 5)	−0.65	−1.74 to 0.30	.18
2020-01-01 cohort only (item 3)	−0.03	−1.97 to 1.96	
BY2-BY3 only in step 1 (item 6)	−0.02	−1.98 to 1.97	
No measurement-error model (item 7)	−0.64	−1.51 to 0.23	
Complete cases (item 8)	−0.06	−2.04 to 1.95	
Unadjusted county spending (item 9)	−0.04	−1.99 to 1.93	
Higher BASIC level (C-E)			
Confirmatory	1.34	0.38 to 2.31	
Firth logistic regression (item 5)	1.55	0.53 to 2.70	.006
2020-01-01 cohort only (item 3)	1.57	0.42 to 2.76	
BY2-BY3 only in step 1 (item 6)	1.33	0.41 to 2.28	
No measurement-error model (item 7)	1.26	0.33 to 2.23	
Complete cases (item 8)	1.34	0.41 to 2.32	
Unadjusted county spending (item 9)	1.52	0.61 to 2.47	

Log-odds differences per 2 SD (BASIC level: C-E vs A-B). The Firth regression includes the core predictors and cohort, with profile-likelihood intervals and a direct-simulation P value; it would have decided H3 only if the Bayesian model had failed its convergence checks.

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eTable 5. Convergence and predictive checks of the decision model (H3)

Check	Result
Largest R-hat (Normal / Student-t prior)	1.001 / 1.001
Smallest bulk effective sample size	6,661 / 6,179
Divergent transitions	0 / 0
Fits needed (registered maximum: 2)	1 / 2
Prior predictive share renewing early, median (95% range)	17% (1% to 70%)
Observed early renewers (posterior predictive 95% range)	24 (14 to 37)
Top predicted decile: observed vs predicted share	57% vs 45%
Direct-simulation refits not converged (pull / own / BASIC level)	0.34% / 0.38% / 0.43%
Overall test: largest absolute core coefficient (reduced sampler settings)	1.35, P=.004

eTable 6. Conventional (frequentist) counterparts

Hypothesis	Method	Estimate	95% CI	P
H2b, regional pull	Welch t test	+3.2	1.1 to 5.3	.004
	OLS, HC2 errors, cohort	+3.3	1.3 to 5.4	.001
	OLS, HC2 errors, adjusted	+3.0	0.9 to 5.1	.005
H2a, own-spending component	Welch t test	−2.1	−3.9 to −0.2	.03
	OLS, HC2 errors, cohort	−2.1	−3.8 to −0.4	.02
	OLS, HC2 errors, adjusted	−1.6	−3.3 to 0.1	.06
H1, savings rate PY2020	Welch t test	+1.8	0.4 to 3.3	.01
	OLS, HC2 errors, cohort	+1.9	0.5 to 3.3	.009
	OLS, HC2 errors, adjusted	+1.4	−0.1 to 2.8	.06
H4, event study (TWFE)	PY2020 vs PY2021	−0.7	−4.8 to 3.4	.73
	PY2022 vs PY2021	−0.4	−2.8 to 2.0	.75
	PY2023 vs PY2021	−2.6	−7.4 to 2.2	.29
	PY2024 vs PY2021	−3.1	−7.9 to 1.7	.20

Reported only; not used for any decision. Differences are early renewers minus stayers (pp; H1 in pp of savings rate). Adjusted OLS adds track, level and size to cohort. Event-study coefficients are log points (0.01 in log spending) relative to PY2021, from two-way fixed effects with errors clustered by ACO. TWFE, two-way fixed effects.

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eTable 7. Agreement of registered and conventional verdicts

Claim	Registered method	Conventional method	Agree
H2b	Supported	Supported	Yes
H2a	Contradicted	Contradicted	Yes
H3 regional pull	Supported	Supported	Yes
H3 own-spending component	Undetermined	Undetermined	Yes
H3 higher BASIC level	Supported	Supported	Yes
H4	Undetermined	Undetermined	Yes

For H2a and H2b the registered conventional counterpart is OLS with HC2 errors; for H3, maximum-likelihood logistic regression; for H4, two-way fixed effects. 'Undetermined' for H4 is the registered label for 'no detectable difference'.

eTable 8. Spending growth after early renewal (H4), with the pre-trend and robustness estimates

Estimate	Early / comparison, No.	Difference, %	95% CrI	*P*
H4, difference-in-differences	**29 / 276**	**−0.4**	**−4.0 to 2.5**	**.78**
Pre-trend, PY2020 to PY2021	29 / 276	+0.7	−2.3 to 4.9	.58
Trend-adjusted	29 / 276	−4.0	−12.0 to 2.7	.14
Risk-adjusted outcome	29 / 276	+0.3	−2.0 to 2.5	.85
Balanced panel (all 5 years)	25 / 162	−1.6	−5.6 to 1.6	.30
Propensity-weighted	305 total	−0.5	NA	.67
2020-01-01 cohort only	18 / 118	−2.9	−8.3 to 1.3	.13
Two-way fixed effects (conventional; 95% CI)	306 ACOs	−1.5	−5.4 to 2.4	.44

Abbreviations: CrI, credible interval; NA, not available (the propensity-weighted estimate has a permutation P value but no interval; its largest weight is 0.83); PY, performance year. The outcome is the change in log per-capita spending from PY2020-PY2021 to PY2022-PY2024, shown as a percent difference in spending growth, early renewers minus ACOs not yet renewed. A stayer's years after it renewed into a 2023 or 2024 agreement are excluded. Equivalence condition: the pre-trend bound scaled to the distance between period midpoints (2.41 years) is 11.5 log points, above the absolute H4 estimate of 0.4 log points (1 log point = 0.01 in log spending, about 1%), so the condition does not hold and H4 is not read as an effect. The two-way fixed-effects model has one more ACO: the stayer with no post-period spending value keeps its pre-period years.

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eTable 9. Attrition from the H4 spending panel, by group and year

Group and status	PY2022	PY2023	PY2024
Early renewers (29): included	29	28	25
Early renewers (29): left the program	0	1	1
Early renewers (29): renewed (no longer 'not yet renewed')	0	0	3
Stayers (277): included	276	222	162
Stayers (277): left the program	0	46	72
Stayers (277): renewed (no longer 'not yet renewed')	0	8	42
Stayers (277): no spending value	1	1	1

Required by the registered plan (section 4.5). PY2020 and PY2021 include all 29 early renewers and 277 stayers. Follow-up: a mean of 2.83 included post-period years per early renewer and 2.39 per comparison ACO (of 3).

eTable 10. Exploratory analyses

Analysis	Estimate	95% CrI
H2b, early renewers minus stayers (pp of benchmark)		
Regional pull (confirmatory)	+3.4	1.7 to 5.4
Pull weighted by phase-in weight w	+1.6	0.7 to 2.6
Capped regional component	+1.6	0.9 to 2.2
H3 coefficient on the pull measure (log-odds)		
Regional pull (confirmatory)	+1.34	0.44 to 2.30
Pull weighted by phase-in weight w	+1.20	0.40 to 2.04
Capped regional component	+1.49	0.57 to 2.45
Census region added	+1.43	0.46 to 2.41
Census region in H3 (log-odds vs Northeast)		
Midwest vs Northeast	+0.35	−0.85 to 1.50
South vs Northeast	−0.12	−1.12 to 0.89
West vs Northeast	+0.09	−1.21 to 1.38

Labeled exploratory in the registered plan (section 5.6): no predictions and no verdicts. 2 of the 334 ACOs have no census region; the region model's sample is 206 ACOs (24 early renewers), as in H3.

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eTable 11. The 2 early renewers already in ENHANCED

Variable	ACO 1 (2020-01 cohort)	ACO 2 (2019-07 cohort)
Savings rate, PY2020, pp	3.4 (38)	5.6 (65)
Regional pull, pp	0.0 (26)	+2.3 (61)
Own-spending component, pp	−3.8 (41)	−4.7 (29)
Regional component, pp	+1.9 (44)	+3.0 (68)

The 2 early renewers already in ENHANCED in PY2021, for whom renewal could not raise the sharing rate, reported individually against the stayer distribution as the plan requires (section 5.6). Values with the stayers' percentile in parentheses. Exploratory; no test.

eTable 12. Per-capita spending by agreement year

Agreement start	Agreement year	Scheduled benchmark year	Group	ACOs, No.	Mean spending, $
2020	1	No	Other ACOs	135	10,609
2020	1	No	Early renewers	18	9,898
2020	2	No	Other ACOs	127	11,430
2020	2	No	Early renewers	18	10,880
2020	3	Yes	Other ACOs	118	11,860
2020	4	Yes	Other ACOs	96	12,450
2020	5	Yes	Other ACOs	73	13,206
2022	1	No	Other ACOs	188	11,288
2022	1	No	Early renewers	18	10,931
2022	2	No	Other ACOs	164	11,912
2022	2	No	Early renewers	17	11,205
2022	3	Yes	Other ACOs	120	12,762
2022	3	Yes	Early renewers	15	11,892

Mean per-capita spending (US dollars) of assigned beneficiaries by agreement start year and agreement year (performance year - start year + 1), PY2020-PY2024 (plan section 5.6). Agreement years 3-5 are the scheduled benchmark years of the next agreement; early renewers out of a 2020 agreement are shown separately. Benchmark-year status cannot be separated from tenure here, so this is description only.

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eTable 13. Post hoc checks of the expected gain against CMS's published 2022 benchmarks, and of the regional pull against past savings

Check	ACOs, No.	Spearman ρ	Median absolute error	Mean error	Same sign	Mean rebuilt / actual
Regular renewers: rebuilt vs actual change	101	0.61	1.7	−0.3	80%	−0.9 / −0.6
Early renewers: rebuilt gain vs realized reset	26	0.60	2.1	−0.2	92%	−1.6 / −1.4

Post hoc and exploratory; not registered (stharrold.github.io, analysis/mssp-benchmark-ratchet/posthoc/). Changes are 100 x (2022 historical benchmark - PY2021 updated benchmark) / PY2021 updated benchmark, in percentage points of the PY2021 benchmark, both in 2021 dollars: rebuilt from public files, or actual as CMS published it in the PY2022 results file. Error = rebuilt - actual. Realized reset, all 29 early renewers: mean −1.3, median −1.5, 66% below the PY2021 benchmark. Updated benchmark PY2021 to PY2022 (includes the annual update for both groups): early renewers +3.1% and stayers +2.9% on average; within-cohort difference 0.0 points (95% CrI, −2.0 to 2.2; P=.97). Regional pull and past savings (267 early renewers and stayers with the variable): Spearman ρ 0.45 with the PY2020 savings rate, 0.23 with the PY2021 savings rate and −0.25 with the own-spending component (stayers alone: 0.42, 0.23 and −0.25).

eMethods. Registration record and deviations

eTable 14. Registration record

Item	Value
Registry	OSF Registries, osf.io/c53w9, doi:10.17605/OSF.IO/C53W9
Registered	October 26, 2026, 13:22:16 UTC; public, no embargo
Plan	`PREREGISTRATION.md` version 0.6, SHA-256 `d0e374c8b493eb2f2fa7da1c0d8cbe46745a601233f11a81d2b2be73c402e8f2`
Code	commit `a4064f5`, archive SHA-256 `4c98d76a586151ab608851c69217a5ddb6b394201643a3de89a839bb1f659cd4`
Data	40 CMS public use files downloaded October 22, 2026, each with its SHA-256 hash in `data-snapshot.tsv`; archived on OSF[2]

Outcome lock. Every data file is hashed before it is read. A file whose hash matches a performance-year results file, a county spending file or the county assignment file could not be opened until the registration record existed, and the code had no bypass. Before registration, only the enrollment

files, the CMS data dictionaries, the regulation text and the column names (not the values) of the outcome files were used.

Order of analysis. The stages ran in the order validate, power, confirm and robustness (concurrently; both depended only on validation and power), then H4, conventional and exploratory. Each stage refused to start until its inputs existed. The robustness results were committed about 17 hours before the confirmatory results and were read only together with them, primary test first.

Deviations. All 7 stages ran with code identical to the registered commit. Two files changed after the last stage and produced no result: a reporting script for the online dashboard, which had shown PY2021 savings rates 100 times too large, and the lockfile `uv.lock`, which took security updates to 2 packages (filelock and virtualenv), neither of which enters an estimate. The lockfile is a registered file, so its change is logged as the one deviation; rerunning the analysis as registered uses the lockfile at the registered commit. One sentence in the plan's background (section 1.1) is imprecise: ENHANCED shares 75% of savings but up to 75% of losses, and the bar on moving from BASIC to ENHANCED within an agreement period is in the 2018 rule's preamble,[3] not in the regulation's text. Nothing in the analysis depends on it.

Reproducibility. A rerun from a fresh clone reproduced the validation stage exactly and every deterministic value of the power stage, but not the simulated values exactly: H1's simulated power was 77.6% against the committed 80.4%, and H3's detection shares 15% against 17% (pull) and 15% against 14% (BASIC level). The analysis reads its table without a fixed row order, so seeded random draws fall on ACOs in a different order in each run, and every resampled or simulated value varies within simulation error (about 1.6 points for a detection share from 500 datasets). No verdict is near a threshold at which that would matter.

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